Deployment Precision Calibration

Frozen HYB-MS +10% ensemble • identity-disjoint calibration only

GATE FAILED

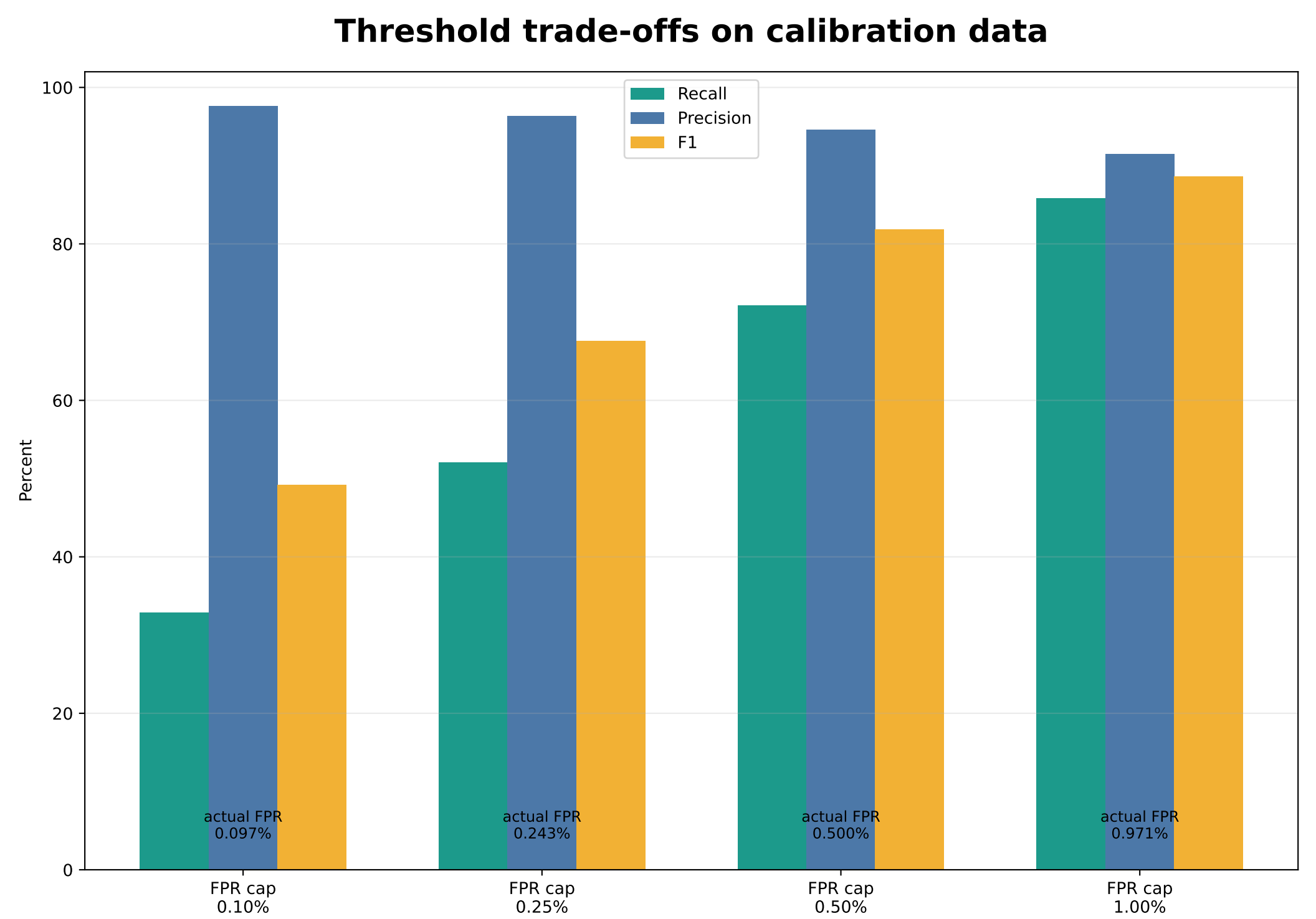
Selected threshold: 0.999411

Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
94.60%	96.30%	52.08%	67.60%	98.10%	92.66%	0.24%	47.92%

Gate: FPR ≤ 0.25% and recall ≥ 80%. Observed: FPR 0.24%, recall 52.08%.

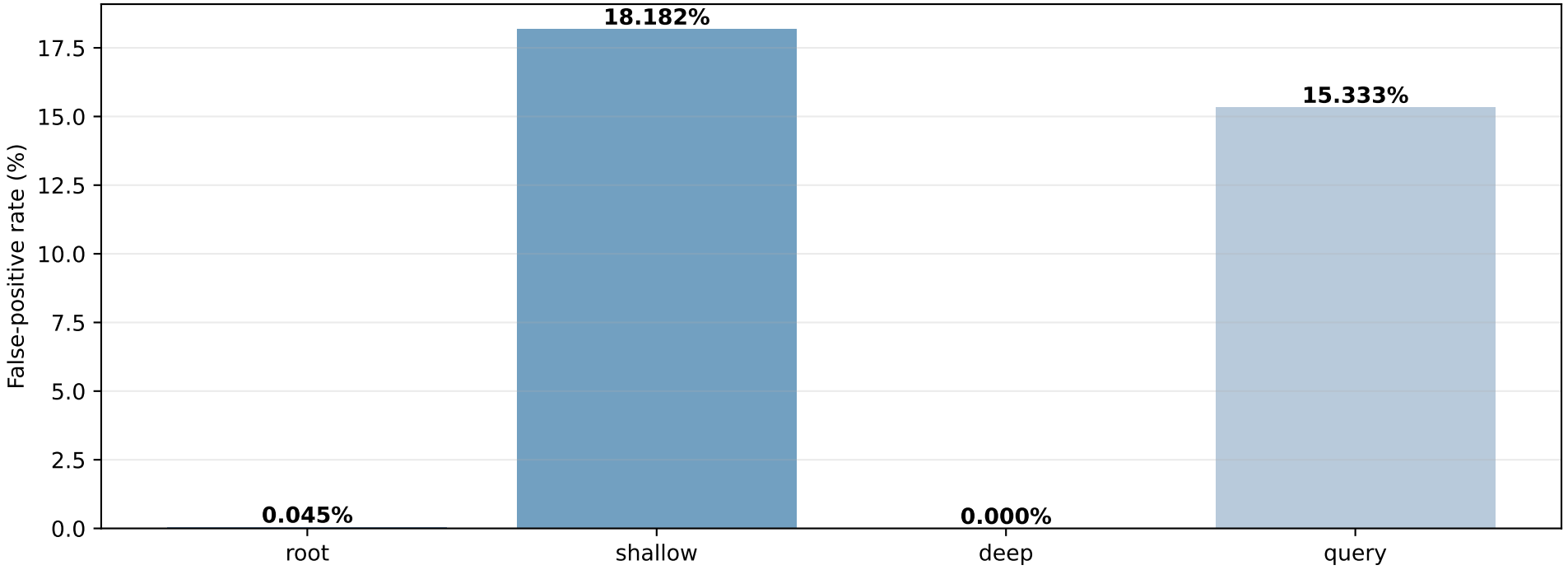
Counts: TN 20,550 • FP 50 • FN 1,198 • TP 1,302

This threshold was chosen without using the final confirmation set. If the gate fails, the model must be retrained using new development data and recalibrated on a separate cohort.



Original threshold 0.999512: recall 50.16%, FPR 0.23%. Selected deployment threshold is 0.999411.

Where False Positives Occur



Feature-pattern counts among false positives

long url: 17 • query string: 23 • none of registered flags: 26 • punycode: 1

These categories identify likely collection or representation weaknesses; they are diagnostic evidence, not permission to tune against the untouched confirmation set.

Operational Interpretation

At 0.1% phishing prevalence	**projected precision 17.68%**
At 1% phishing prevalence	**projected precision 68.43%**
At 5% phishing prevalence	**projected precision 91.87%**
At 10% phishing prevalence	**projected precision 95.97%**

Retraining decision

Gate failed; threshold adjustment alone is insufficient. Retrain with newly collected, structurally diverse legitimate development rows, then recalibrate on a separate unused cohort.

Next locked action

Apply this exact ensemble and threshold once to a newly frozen, identity-disjoint low-prevalence confirmation cohort. Do not revise the threshold after viewing those labels.